

Lesson 5.2: Risk preference metrics

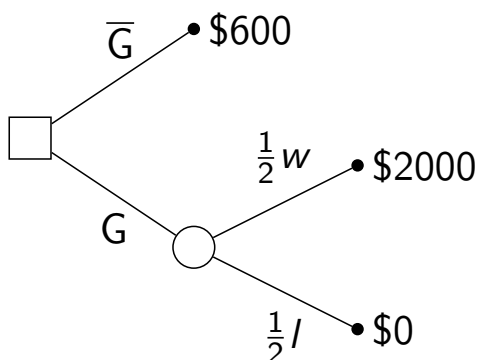
Risk attitudes and utility

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Certainty vs risk

Example (Certainty equivalent)

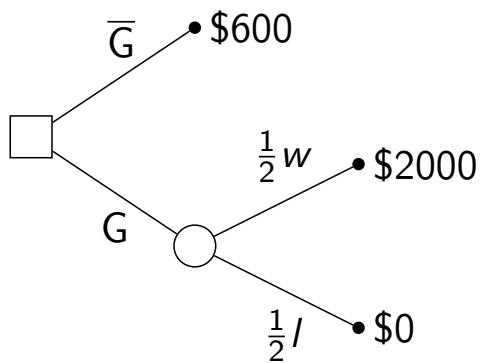
Alice is risk averse; when offered a one-in-two chance of winning \$2000, she would only take the risk for \$600 or less.



- ▶ Alice prefers a certain \$600 to the *lottery*
- ▶ \$600 is Alice's *certainty equivalent* for the lottery:

$$\left[\frac{1}{2} : \$2000 \mid \frac{1}{2} : \$0 \right]$$

Risk premium



- ▶ \$600 is \$400 less than the *fair value* of the lottery (\$1000)
- ▶ her *risk premium* is \$400
- ▶ her risk premium quantifies how risk averse she is

Definition (Risk premium)

A decision-maker's *risk premium* is the difference between the fair value of an uncertain alternative and the decision-maker's certainty equivalent for it.

Risk attitudes: reformulated

- ▶ Risk averse = positive risk premium
Alice will accept less than fair value of uncertain alternative to avoid risk (in this case up to \$400 less)
- ▶ Risk seeking = negative risk premium
A risk seeking Alice would pay more than fair value of uncertain alternative for a chance to win
- ▶ Risk neutral = zero risk premium